



TROCKERS ONLINE REVISION

General Certificate of Education Advanced Level

MATHEMATICS

6082/1

PAPER 1

NOVEMBER EXAMINATION PRACTICE A

2026 SESSION

3 hours

Additional materials:

**Answer Paper Non-programmable calculator
Graph Paper List of formulae**

TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name in the spaces provided on the answer sheet/answer booklet.

Answer all questions in Section A and any five questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Non-programmable calculators may be used.

You are reminded of the need for clear presentations in your answers.

This question paper consists of 8 printed pages.

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[Turn Over

Section A: 60 Answer all questions

1. Given that

$$\frac{x^2 + 5x}{2} + \frac{42}{x^2 + 5x} = 10.$$

Using an appropriate substitution, determine the values of x .

[4]

2. Find the independent term in the expansion of $\left(2x + \frac{3}{x}\right)^4$.

[3]

3. A sequence is defined by

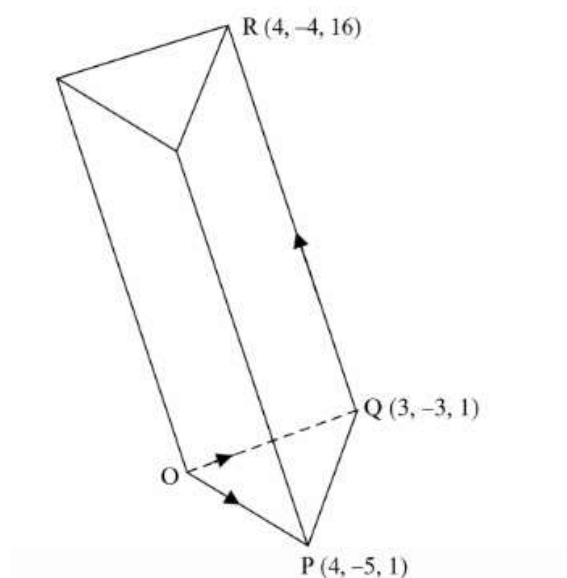
$$u_n = 2 + \left(\frac{1}{2}\right)^n$$

Find the first three terms of the sequence and determine the behaviour of the sequence as $n \rightarrow \infty$.

[3]

4. The points O, P, Q and R have position vectors

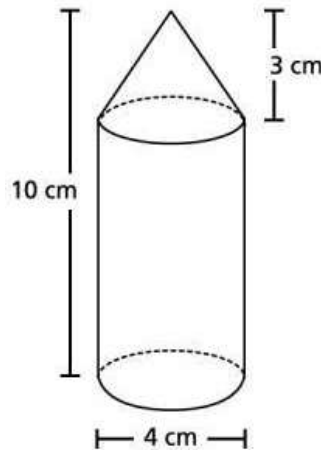
$$O = (0,0,0), P = (4, -5, 1), Q = (3, -3, 1), R = (4, -4, 16).$$



Find angle PRQ .

[4]

5. Find the coordinates of the centre and the exact radius of the circle given by $x^2 + y^2 - 4x + 6y - 77 = 0$. [3]
6. Points A , B , and C have coordinates $A(0,0)$, $B\left(\frac{2}{5}, 0\right)$, $C\left(\frac{4}{25}, 1\right)$.
The point M is the midpoint of AB . Find the
(a) coordinates of M ,
(b) equation of the line MC . [4]
7. A circle is divided into 6 sectors such that the angles of the sectors are in arithmetic progression. Given that the largest angle is 5 times the smallest angle, find the angle of the largest sector. [4]
8. A model silo of height 10 cm is made by placing a cone on top of a cylinder. The cylinder has a diameter of 4 cm, and the cone, which has the same base radius as the cylinder, has a height of 3 cm. The diameter is measured with a possible error of 0.15 cm.



Cylinder: $V = \pi r^2 h$

Cone: $V = \frac{1}{3} \pi r^2 h$.

Calculate the approximate percentage error in the calculated volume of the silo. [4]

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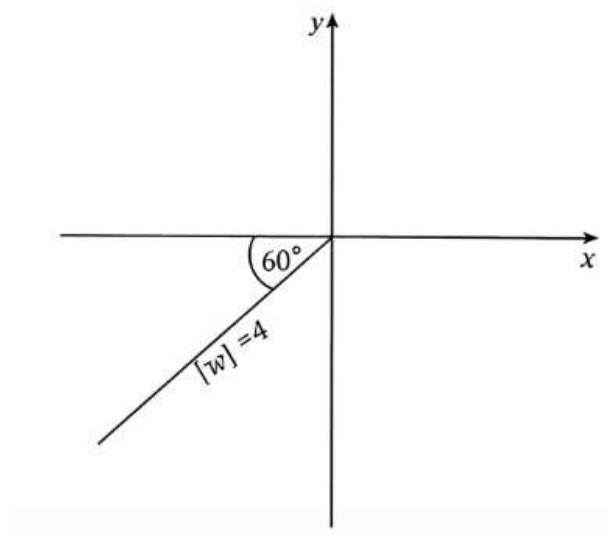
9. Use the substitution $u = \cos x + 2$ to evaluate

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{(\cos x + 2)^3} dx. \quad [5]$$

10. The function is defined by $y = x^2 e^{-x}$.

Find the coordinates of the stationary points. [4]

11. A complex number w has modulus 4 and lies in the third quadrant, as shown on the Argand diagram below.



Express w in the form $a + ib$, where a and b are exact real numbers. [3]

12. (a) A function y satisfies the differential equation

$$\cos^2 x \frac{dy}{dx} = 1.$$

Given that $y = 0$ when $x = 0$,

(i) Solve the differential equation, expressing y in terms of x . [4]

(ii) Sketch the graph of the solution for $0 < x < \frac{\pi}{2}$. [2]

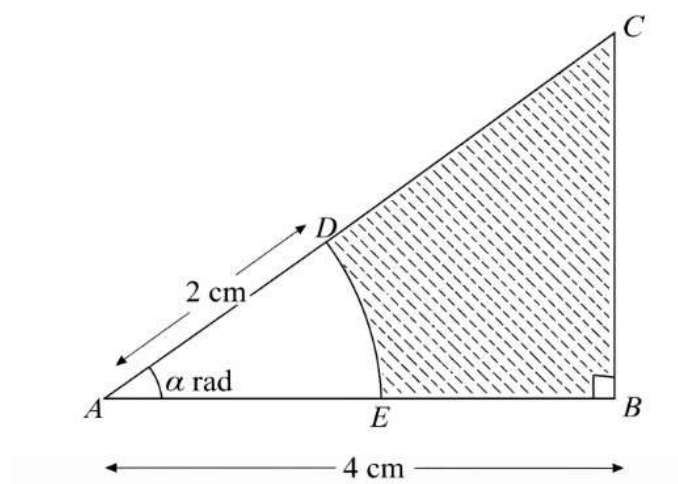
13. A variable p is inversely proportional to $(2\sin q + 3)^2$.

Given that $p = 2$ when $q = \frac{\pi}{6}$, find the constant of proportionality.

Find the value(s) of q , where $0 \leq q \leq 2\pi$, when $p = 8$. [5]

14. Given that $y = x^2 \sin 2x$, find the Maclaurin series expansion for y , in ascending powers of x , up to and including the term in x^2 . [4]

15. The diagram below shows triangle ABC in which AB is perpendicular to BC . The length of AB is 4 cm and angle $CAB = \alpha$ radians.



An arc DE , with centre A and radius 2 cm, meets AC at D and AB at E . Find, in terms of α , the perimeter of the shaded region. [4]

[Turn Over]

Section B: Answer any 5 questions.

16. Given that $\cos x + x \cos x - x = 0$ for $0 \leq x \leq \pi$,

(a) Express the equation in the form $\cos x = f(x)$, where $f(x)$ is a function of x . [2]

(b) Let $g(x) = \cos x + x \cos x - x$.

Show that $g(x)$ has a root in the interval $[0.5, 1.1]$. [3]

(c) Sketch the graphs of $y = \cos x$ and $y = f(x)$ for $0 \leq x \leq \pi$, and hence determine the number of roots of the equation in this interval. [3]

(d) Using $x_0 = 1$ as the initial value, apply the Newton–Raphson method twice to find the root correct to 3 decimal places. [4]

17. (a) On an Argand diagram, shade the region satisfying

$$|z - (2 + i)| \leq 3 \text{ and } \arg(z - 1) < \frac{\pi}{4} \quad [4]$$

(b) Given that $z = 2 \left[\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right]$, find the modulus and argument of z^3 . [2]

(c) Solve the equation $z^4 + 16i = 0$, giving your answers correct to 1 decimal place in the form $a + ib$. [6]

18. The curve is given parametrically by

$$x = t^2 + 1, y = t^3 - t$$

(a) Find $\frac{dy}{dx}$. [3]

(b) Find the coordinates of the stationary points. [4]

(c) Find the equation of the normal to the curve at the point where $t = 2$, giving your answer in the form $Ax + By = C$. [5]

19. The line L has a vector equation $\mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$.

(a) Find the angle between the line L and the plane $2x - y + 2z = 5$. [4]

Points A , B , and C have position vectors

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$$

(b) Find the shortest distance from point B to the line L . [4]

(c) Show that points A , B , and C are collinear. [2]

(d) The point D has a position vector $\mathbf{d} = \begin{pmatrix} 2 \\ 3 \\ k \end{pmatrix}$

Find the value of k such that $\mathbf{AD} \perp \mathbf{AC}$. [2]

20. (a) Given the matrix

$$A = \begin{pmatrix} 2 & 1 & -1 \\ 1 & 3 & 2 \\ 0 & -2 & 1 \end{pmatrix}$$

i) Find $\det(A)$. [3]

ii) Find A^{-1} . [5]

(b) Hence, solve the system of equations:

$$\begin{aligned} 2x + y - z &= 3 \\ x + 3y + 2z &= 7 \\ -2y + z &= -1 \end{aligned}$$

[4]

21. (a) Express $5\sin x + 12\cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and

$$0 < \alpha < \frac{\pi}{2}. \quad [3]$$

(b) Hence,

(i) find the maximum and minimum values of

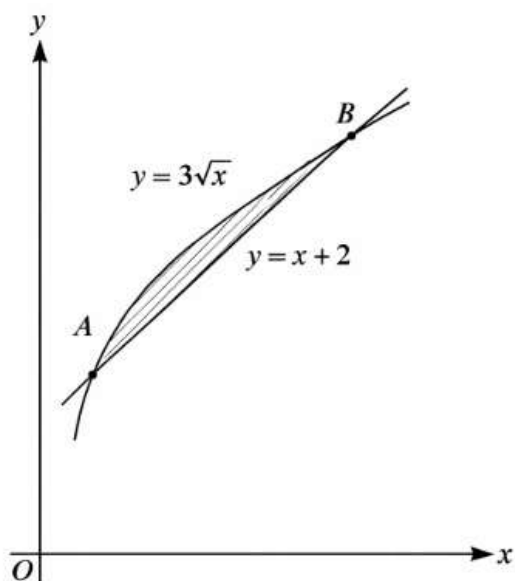
$$\frac{1}{15 + 5\sin x + 12\cos x} \quad [2]$$

(ii) solve the equation $5\sin x + 12\cos x = 6$ for $0^\circ \leq x \leq 360^\circ$. [4]

(iii) sketch the graph of $y = 15 - 5\sin x - 12\cos x$ for $0^\circ \leq x \leq 360^\circ$,
clearly showing the maximum and minimum values. [3]

[Turn Over]

22. (a) The diagram shows parts of the graphs of $y = x + 2$ and $y = 3\sqrt{x}$ which intersect at points A and B .



Calculate the area of the shaded region enclosed between the two curves. [5]

- (b) (i) Use the binomial expansion to show that

$$\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2 \quad [4]$$

- (ii) Substitute $x = \frac{1}{11}$ into

$$\sqrt{\frac{1+4x}{1-x}} \approx 1 + \frac{5}{2}x - \frac{5}{8}x^2,$$

to obtain an approximation for $\sqrt{6}$. Give your answer as a fraction in its simplest form. [3]

It is not how you start the race or where you are during the race, but it is how you cross the finish line that will matter: 2nd Timothy 4 vs. 6-7.

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